THE UNITED REPUBLIC OF TANZANIA
ORGANIZATION OF HEADS OF NON-GOVERNMENT SECONDARY SCHOOLS (OHONGSS)
KIGOMA REGION

FORM FOUR MOCK EXAMINATION
(ACTUAL PRACTICAL)

031/2 **PHYSICS 2A**
Time: 2:30 Hours **YEAR: 2024**

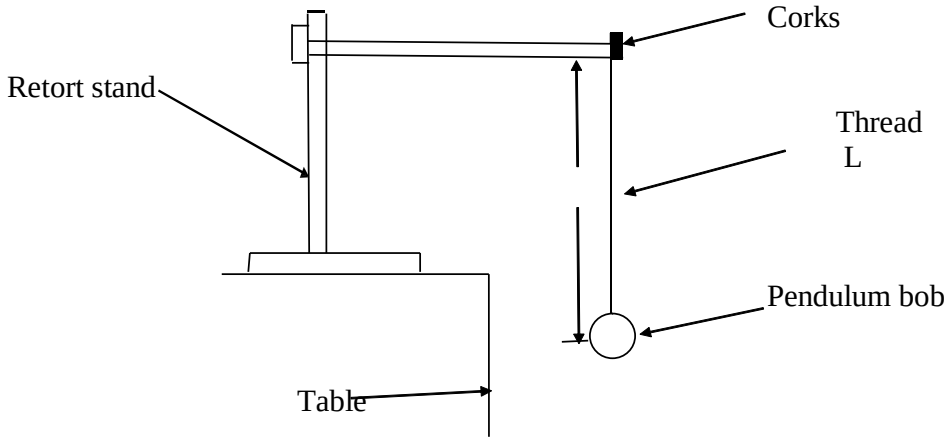
Instructions

1. This paper consists of two (2) questions.
2. Answer all questions.
3. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
4. Each question carries twenty five (25) marks.
5. Non programmable calculators may be used.
6. All writings should be in blue or black ink except for graphs and diagrams which must be drawn with pencil.
7. You may use the following constants;
   (i) Acceleration due to gravity g = 10 N/kg

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1. You are provided with a pendulum bob, stop watch, thread, and retort stand with clamp. Proceed

as follows. Set up the apparatus as shown in the figure below.



a) Adjust the thread of length (L) of the pendulum bob to be 140 cm. Set the bob into oscillations by displacing it to a small and release it. Record the time for 10 oscillations.

b) Repeat (a) above for length of pendulum L = 100cm, 80cm, 60cm and 40cm. Record your results in a suitable table including periodic T and T^2.

c) Plot the graph of T^2 against L

d) Determine the slope of your graph.

e) From the graph , state the relationship between T^2 and L

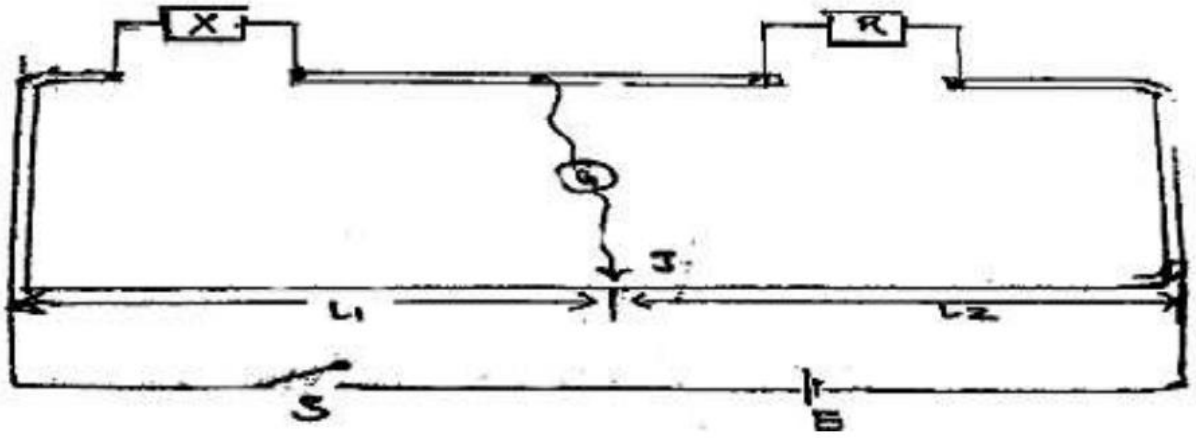
f) Using the graph find the time taken by the bob to oscillate when the length of the pendulum is L = 50 cm

g) Given that $\frac{1}{L} = \frac{2\pi^2}{\frac{1}{g^2 T}}$, calculate the acceleration due to gravity.

h) Suggest the aim of this experiment.

i) State any two sources of errors. (**25 Marks**)

2. You are provided with a resistance box R, a dry cell E, a switch S, a jockey J and a zero centered galvanometer. Your are required to determine the value of the unknown resistance X by means of a meter bridge as follows:



(a) Set up the circuit as shown above

(b) Set R =1Ω, close the switch, slide the jockey over the meter bridge wire until the galvanometer reads zero. Read and record the length. Also read and record the corresponding length.

(c) Repeat the procedures in 2 (b) for values of R = 2Ω, 3Ω, 5Ω and 7Ω and record the values for and in each case.

QUESTIONS

(i) Tabulate your results including the values of $\frac{L_2}{L_1}$

(ii) Plot a graph of R (Ω) against $\frac{L_2}{L_1}$

(iii) Find the slope S of the graph

(iv) Determine the value of the unknown resistance X by showing clearly how you arrive at your answer (**25 Marks**).